

Stage 1: Laptop CPU Benchmark Results

Testing Environment & Parameters:

Hardware: Laptop CPU (x86 architecture)

Resolution: 640 × 480 (307,200 pixels)

Max Iterations: 30

Total Newton Iterations: 1,743,986

Performance Comparison: C++ vs Python (Single Frame)

Implementation	Latency (ms)	Frame Rate (FPS)	Iteration Rate (Mit/s)
C++ (g++ -O2)	54.58	18.32	31.95
Python (Reference)	1403.90	0.71	1.24

Note: The Mit/s (Million iterations per second) metric represents the raw computational throughput and serves as the fairest measure for cross-platform performance comparison.